Senior Design Meeting 10/8/2026

SDMAY27-40

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IOWA STATE UNIVERSITY

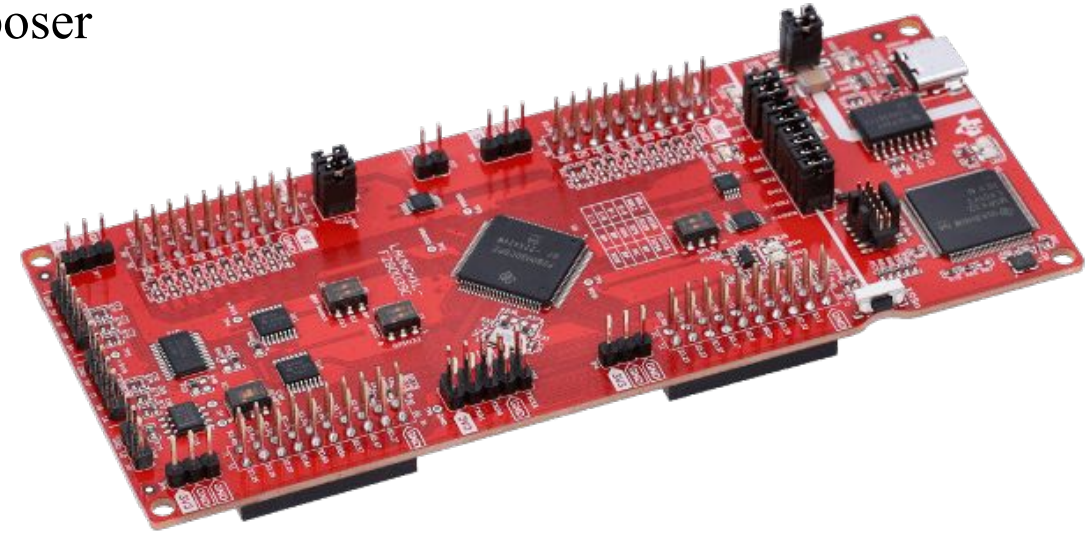
Project Description

The open-source controls microcontroller is an MCU targeted towards embedded systems with special functionality for control loops (i.e. PID). With embedded systems requiring fast response times, like FPV UAVs, the microcontroller will have peripherals to interface with other devices over common communication protocols, hardware acceleration for PID algorithms to meet loop frequency requirements, and interfaces over PWM to output to motors. This gives applications fast response times, while still being able to have low processor utilization and minimize its SWaP (Size, Weight, Power).



Benefits/Niche

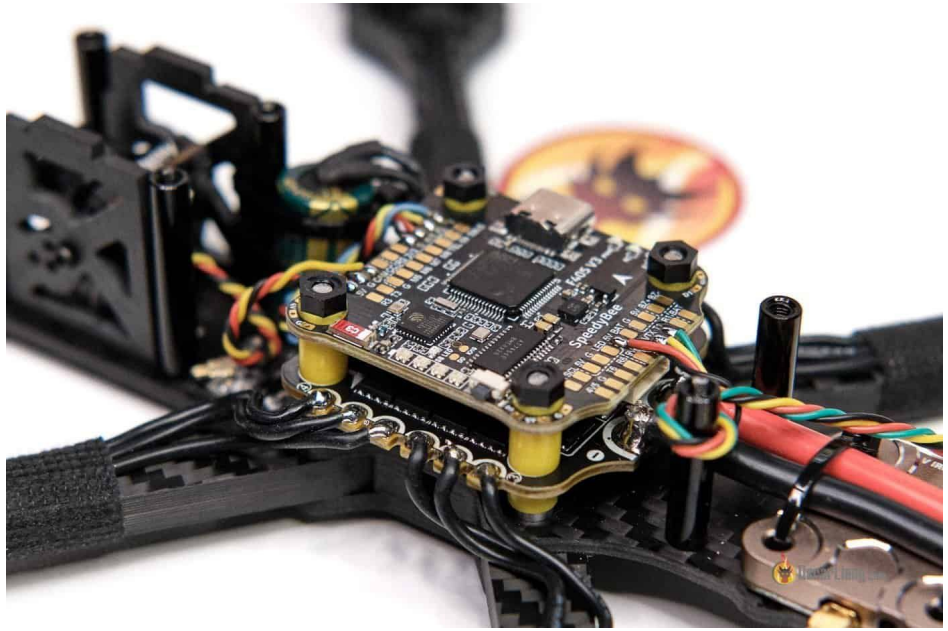
- A specialized ASIC reduces space + energy constraints (SWAP)
- Hardware driven control can achieve quick response times.
- Hardware parallelization allows for less processor utilization, freeing the processor for alternative tasks
- Most similar alternative is the TI C2000 series MCUs, which has some architectural oddities, is limited to Code Composer Studio, and is closed-source



Example Use Case (UAV QuadCopter, Bitcraze Crazyflie 2.1)

- Weighs 29g. Max 44g with payload.
- I2C/SPI for sensors. PWM for motors. UART for telemetry.
- BetaFlight can run PID Loops at 4-8 kHz (latency of 125-250us)
- Board portion has 900 mm^2
- Drone batteries of 250-350 mAh
- Max 4A power draw.
- Accelerometers require trig for attitude estimation.
- STM32F407VG requires 176 cycles (~1us) to compute sine





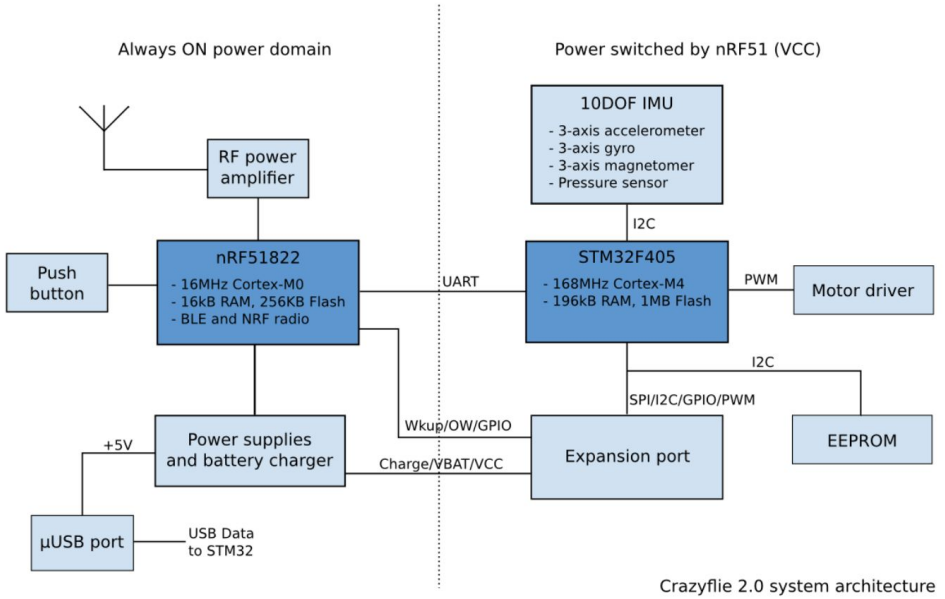
Typical Sensors/Peripherals

- Example target application: Bitcraze CrazyFlight 2.1 Brushless.
- Bosch BMI088 IMU
  - Required Protocols: I2C or SPI
  - Power Consumption: ~17 mW
- Nordic nRF52810 Bluetooth
  - Required Protocols: UART
  - Power Consumption: ~15.18 mW









Crazyflie 2.0 system architecture

Power Breakdown



Category: Micro-Quadcopter
Motor: Crazyflie 2.1 Brushless motors
Battery: 1S 350mAh LiPo

Motor Speed	Power/Motor	All 4 Motors
30%	0.19W	0.78W
50%	0.90W	3.60W
60%	1.56W	6.22W
80%	3.69W	14.75W
100%	7.20W	28.80W



Category: Mini-Quadcopter
Motor: BETAFPV 1102-18000KV Brushless
Battery:1S LiPo (3.7 V nominal)

Motor Speed	Power/Motor	All 4 Motors
30%	0.65W	2.59W
50%	3.00W	12.00W
60%	5.18W	20.74W
80%	12.29W	49.15W
100%	24.00W	96.00W



Category: Standard-Quadcopter 5 inch
Motor: BotSpin 1507 2200KV
Battery: 2S or 3S LiPo

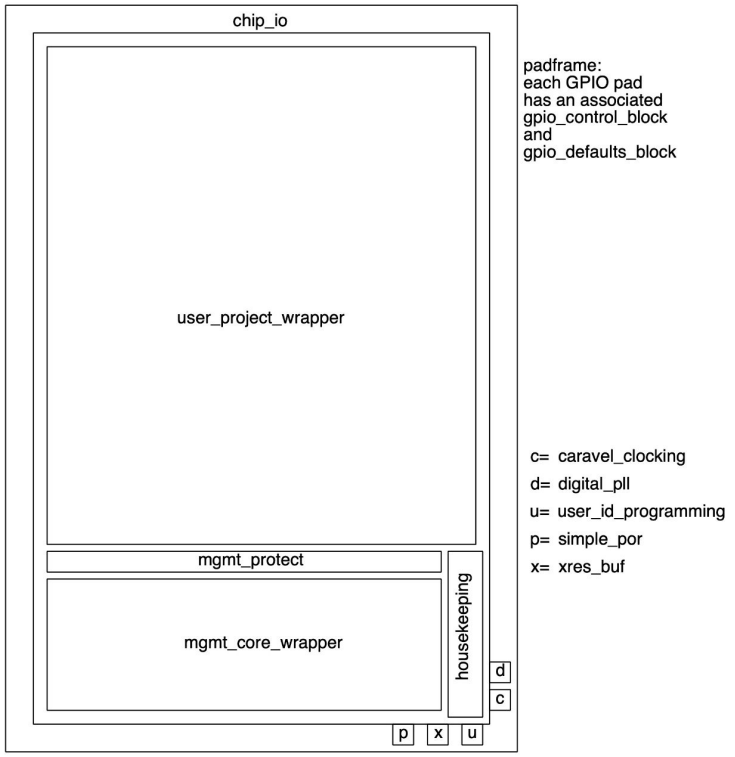
Motor Speed	Power/Motor	All 4 Motors
30%	9.25W	37.00W
50%	25.27W	101.09W
60%	37.48W	149.93W
70%	54.37W	217.50W

Requirements

- Can achieve PID response frequencies of >= 8kHz, in sample case (latency of < 125 us), with accelerometer as input
- Can have up to 3, 2-layer nested PID Loops in parallel (3 axis times an inner+outer loop), 32-bit fixed point.
- Must support up to 4 PWM output channels (one for each motor)
- Support an interface of I2C, SPI, and UART, to have interoperability with Bitcraze CrazyFlight 2.1
- Support fixed-point trig operations of sine, cosine, and atan2 (see math.h) in terms of radians

Considerations/Constraints

- ~¾ of die area and pins (not high concern)
- Data bandwidth between user space and management harness
- On-chip memory must fit within accommodated die area (size dependant on memory technology)
- How to program processor
- Must be manufacturable on the Sky 130nm process
- Tapeout deadline by December + May



Risks

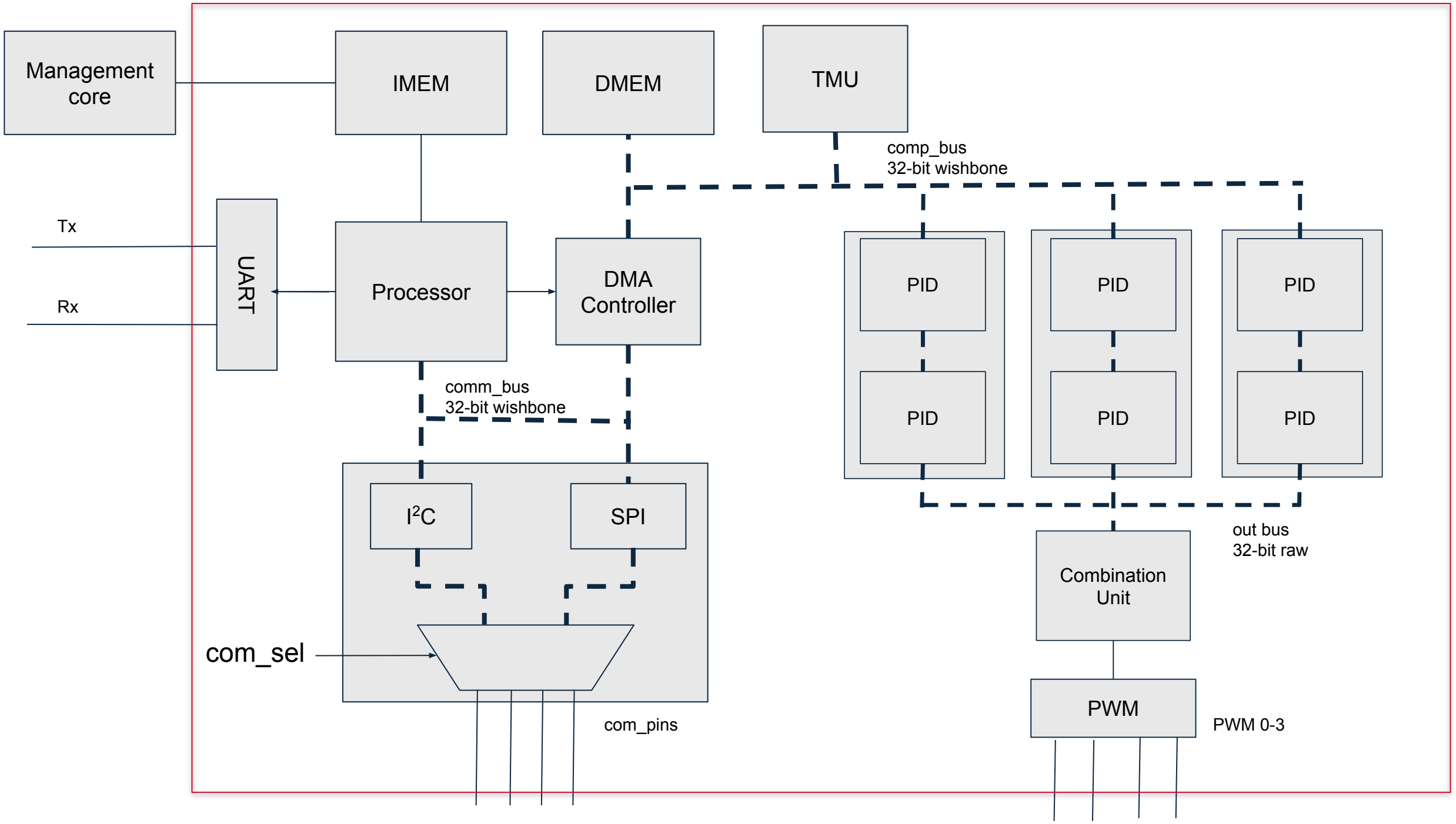
- Chip Foundry going bankrupt
- more I/O or space used by other projects than expected
- Timing failures/tight timings resulting in non-deterministic functionality.





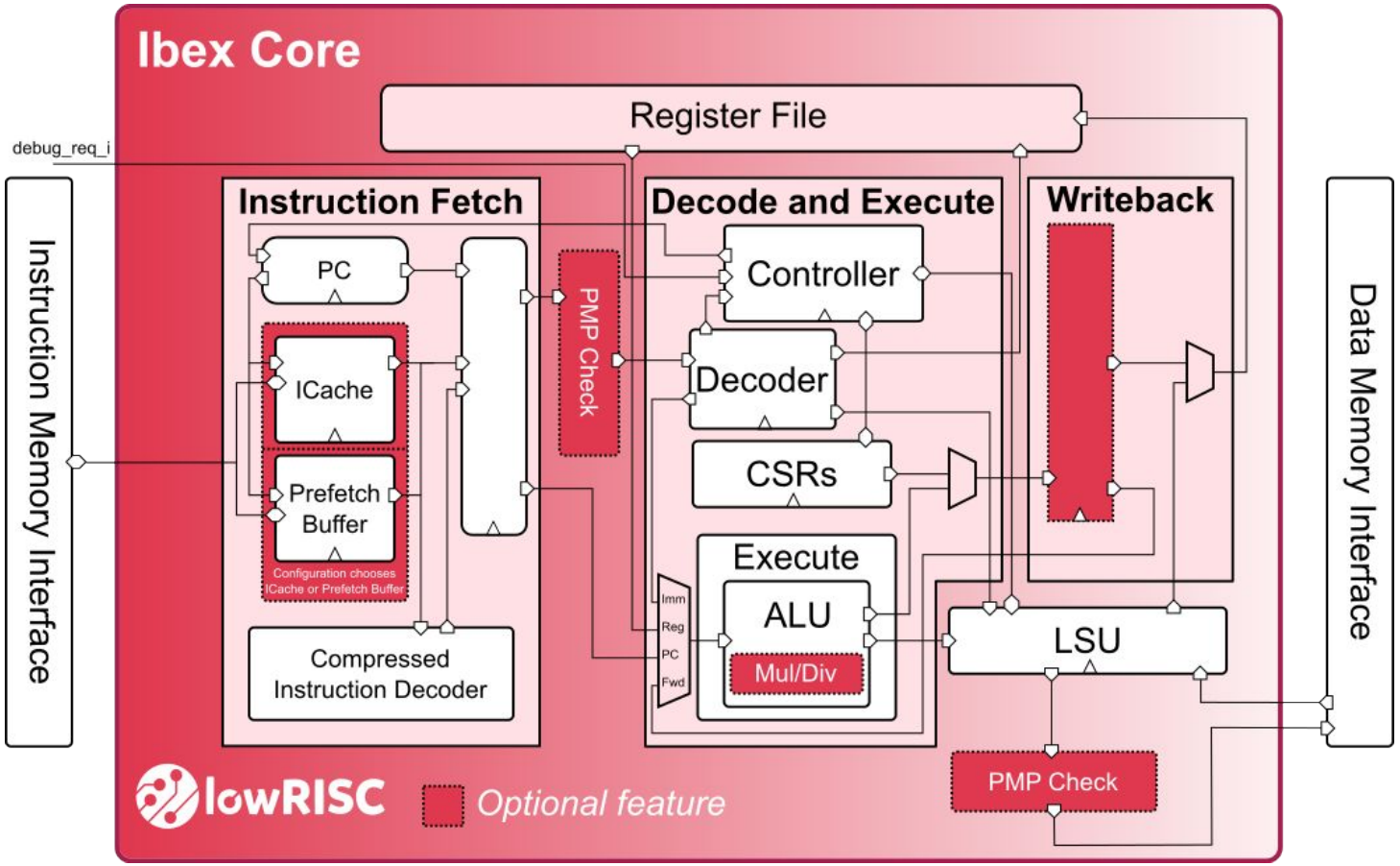
Members’ familiarity with toolflow

Member	Blinky	Flashed to FPGA	Hardened module	Gate level simulations	SVUnit Sims		
Christopher							
David							
Dhruv							
Ian							
Jay							


Management core
IMEM
DMEM
TMU
comp_bus
32-bit wishbone
Tx
Rx
UART
Processor
DMA Controller
PID
PID
PID
PID
PID
PID
comm_bus
32-bit wishbone
I2C
SPI
com_sel
com_pins
out bus
32-bit raw
Combination Unit
PWM
PWM 0-3

Possible RISC-V IP Core: Ibex RISC-V Core

- Extensively Verified.
- Actively maintained by lowRISC (an organization dedicated to supporting open-source silicon)
- Targeted toward embedded applications.
- Has been taped out multiple times.
- Enables simple configuration with various design parameters (for example: can select different types of register file configurations in order to support both FPGA and ASICs)

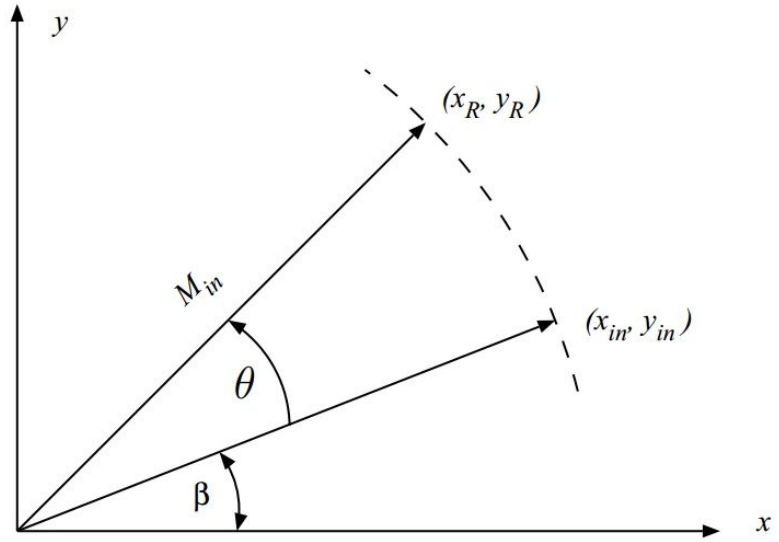


Processor

- Management Processor:
  - Use memory-mapped regions for access to other units
  - Doesn’t require designing processor
  - Easy software support
  - Limited in terms of direct memory access
  - Very slow (bit shifts require up to 31 cycles)
- RISC-V Processor in User Project Wrapper:
  - Use memory-mapped regions for access to other units
  - Instructions can be streamed in by management core
  - Could support basic operations faster than management core
- Custom ISA Processor in User Project Wrapper:
  - Have custom instructions for control/trig
  - Instructions can be streamed in by management core, but may require inline assembly or other solution
  - Could share resources with other functional units for area

Trig Unit

- Lookup table
  - Simple to implement
  - Could use interpolation
  - Memory usage
  - Size scale exponentially with digits of precision
  - Memory could be reduced with quarter-wave symmetry



- CORDIC module
  - Sine and cosine can be calculated simultaneously
  - Implementation requires only shifts and add/sub modules
  - Low memory overhead
  - Size scales linearly with bits of precision
  - Each iteration gives a bit of precision
  - Adjustable precision
  - Takes n cycles

PID Unit

- 2-layer Nested PID units
  - PID units come in pairs
  - Minimizes interconnect of multiple PID units
  - Outputs go into some combination module

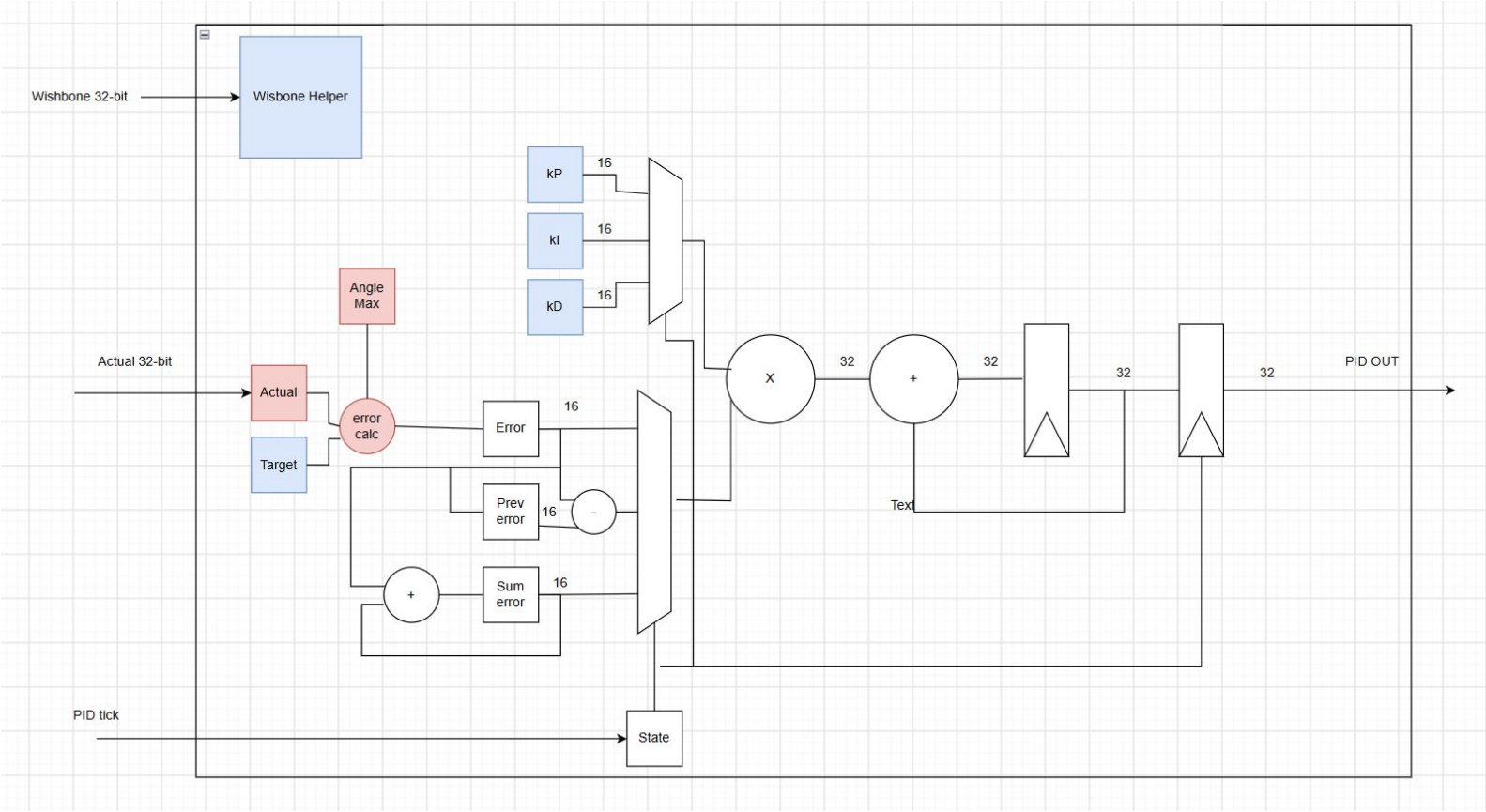
- Full-crossbar PID units
  - PID are connected with each pair
  - Allows high generalizability
  - Overkill for target platform

Peripherals (I2C, UART, SPI, PWM)

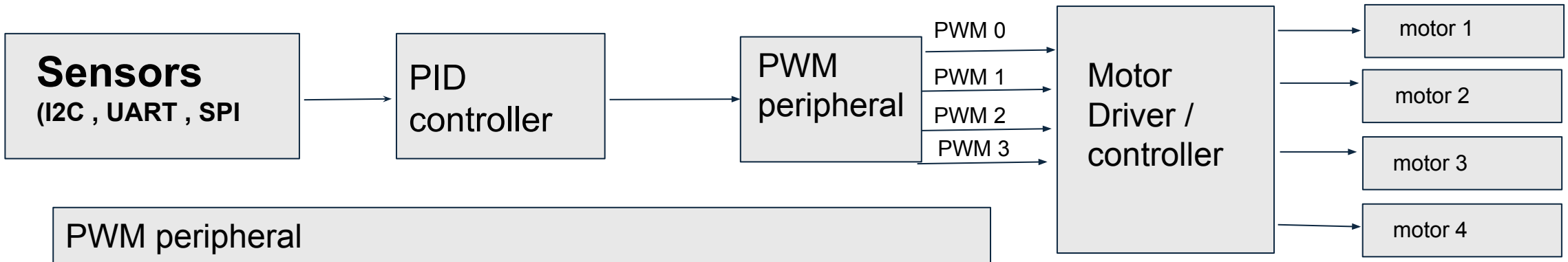
- I2C and SPI are connected via MUX and share pins (these protocols accomplish the same purpose, so they can be toggled interchangeably as need be depending on what peripherals are desired)
- UART needs to be configurable, 8-9 bit with parity (5-7 bit optional?)
- Need PWM pins to control motors, likely will need to use PWM in conjunction with MOSFETs so that any driven motors would get the appropriately-scaled voltage.

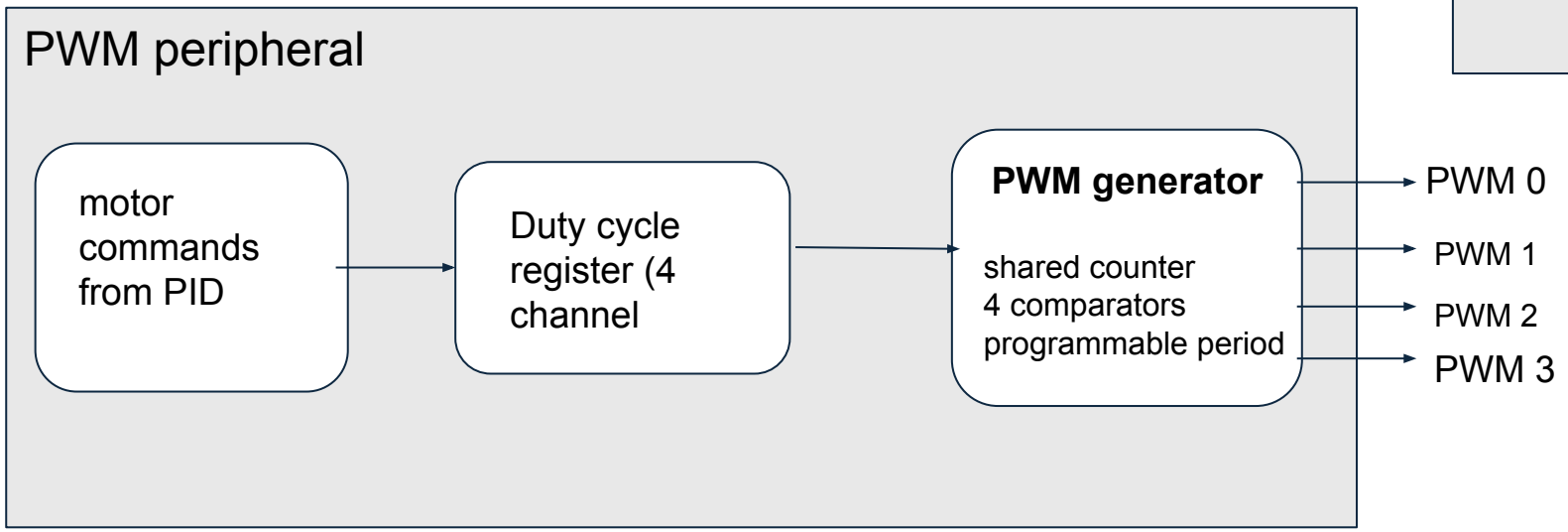
Current PID prototyping

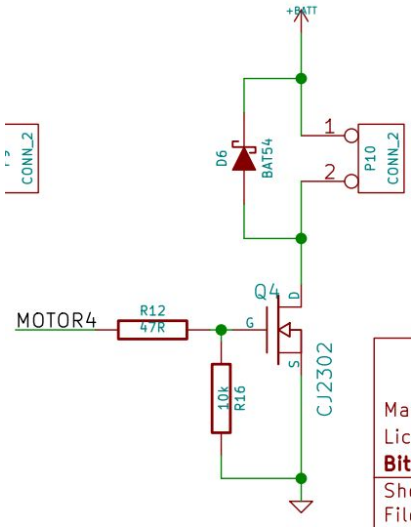
- Got PID calculation path working (though want more intensive testing)
- Set up randomized testing
- Nested case works
- Confirmed interoperability with caravel core
- Getting FPGA testing up by end of week
- Should be able to start hardening



Motor / PWM communication Architecture







Selecting I^2C

I2C IP Provider	OpenCores	PULP	Chip Foundry	OpenTitan/lowRISC
Documentation Link	opencores.org	datasheet.pdf	github.com/chipfoundry	github.com/lowRISC
7-bit slave addressing (required by BMI088)	Yes	Yes	Yes	Yes
repeated start support	Yes	Yes	Yes	Yes
Supports multi-byte read/write	Yes	Yes	Yes	Yes
clock stretching (not supported by BMI088)	No	No	No	Yes
Fast-mode support	No	No	No	Yes
Interrupt flag support	Yes	Yes	Yes	Yes
clock gating	No	Yes	Yes	No

Selecting SPI Cores

SPI IP Provider	OpenCores	PULP	Chip Foundary	lowRISC	OpenTitan
Documentation Link	opencores.org	datasheet.pdf	github.com/chipfoundry	lowrisc.github.io/sonata-system	opentitan.org
Support for dynamic frame size (16-bit required by BMI088)	No	Yes	No	No	Yes
Full duplex communication	---	Yes	---	---	Yes
interrupt flag support	Yes	Yes	Yes	Yes	Yes
configurable CPOL and CPHA	---	Yes	---	---	Yes
variable # of chip select lines	---	No	---	---	Yes
Built-in clock divider	---	Yes	---	---	Yes
Double Transfer Rate support	---	No	---	---	No

Selecting UART Module

IP Provider:	efabless	OpenTitan	OpenCores
Bus Interface:	Wishbone, APB, AHBL	TL-UL	Wishbone
Native SW libraries?	Yes	No	No
Interrupts?	Yes	Yes	Yes
FIFOs/Buffer?	Yes (FIFO)	Yes (Buffer)	Yes (FIFO)
Configuration Options:	5-9 bit data length; 1-2 stop bits; Odd/Even/No Parity/Sticky Parity	8 bit data length; 1 stop bit; Odd/Even/No Parity	5-8 bit data length; 1-2 stop bits; Odd/Even/No Parity
Max speed on Sky130?	83 MHz (Wishbone)	N/A	N/A
Tapeout?	Yes	Yes	?

Selecting PWM

requirements	SKY130 4× PWM	OpenCores PWM	OpenTitan	OpenCores PTC
IP Link				
Bus Interface	Wishbone	Wishbone	TL-UL	Wishbone
4 PWM Outputs	Yes	1 per instance	Yes (6)	1 per instance
Independent Duty Cycles	Yes	Yes	Yes	Yes
Programmable Period	?	Yes	Yes	Yes
Shared Counter	?	No	yes	No
SKY130 Implementation	yes	?	?	?
Wishbone Adapter Needed	No	No	YES	No

Selecting RISC V Processor

	Ibex(lowRISC)	PicoRV32	CV32E40P	VexRiscv
Provider	lowRISC	YosysHQ	OpenHW group	SpinalHDL
Suitable for the Embedded system	Yes	Yes	Yes	Yes
Wishbone compatibility	Adapter needed	Native	Adapter needed	Configurable
Supports fixed point calculations	Yes	Yes	Yes	Yes
Extensive Verification Support	Yes	Limited	Yes	Yes
Multiple documented tapeout	Yes	–	Core lineage	Depends on the implementation
Actively Maintained	Yes	No	Yes	Yes
Separate Instruction/data interfaces	Yes	No	Yes	Configurable
Standard RISCV interrupt support	Yes	No	Yes	Configurable
Small area configuration	Yes	Yes	Less area focused	Yes
Hardware multiplier	Yes	Optional	Yes	Configurable